SUPPLY CHAIN

PERFORMANCE

OPERATIONS

ANALYSIS

WELCOME

Supply Chain

"Network“
(The interconnected system of suppliers, manufacturers, and distributors)

OPERATIONS

"Execution"
(Day-to-day activities of order processing and delivery)

PERFORMANCE

"Efficiency"
(How well the system meets delivery and quality targets)

ANALYSIS

"Insights"
(Data-driven examination to identify improvement opportunities)

Presentation Overview

PROJECT INTRODUCTION
&
PROBLEM STATEMENT

In today's competitive e-commerce landscape, supply chain efficiency is critical for business success. Customer expectations for fast, reliable delivery continue to rise, while operational costs must be controlled.

PROBLEM STATEMENT

- **Delivery Performance Issues:** Inconsistent on-time delivery rates affecting customer satisfaction
- **Revenue at Risk:** Significant portion of revenue threatened by operational inefficiencies
- **Lack of Visibility:** No comprehensive view of performance across shipping modes, departments, and customer segments
- **Resource Allocation:** Unclear which areas need immediate attention and investment
- **Customer Understanding:** Limited insights into customer behavior and profitability patterns

Project Objective

To analyze 4 years of supply chain operations data (2015-2018) and develop interactive dashboards that provide actionable insights for:

- Improving delivery performance and operational efficiency
- Reducing revenue at risk
- Optimizing resource allocation across departments
- Understanding customer segment behavior
- Identifying growth opportunities

OVERVIEW OF SUPPLY CHAIN DATASET

group1=order info						group2= delivery					
Order	order date (DateOrder	shipping date (DateOrder	Order Status	Type	Order Item Quanti	Days for shipping (rea	Days for shipment (schedule	Delivery Status	Late_delivery_ri	Shipping Mo	Cu
77202	1/31/2018	2/3/2018	COMPLETE	DEBIT	1	3	4	Advance shipping	0	Standard Class	
75939	1/13/2018	1/18/2018	PENDING	TRANSFER	1	5	4	Late delivery	1	Standard Class	
75938	1/13/2018	1/17/2018	COMPLETE	CASH	1	4	4	Shipping on time	0	Standard Class	
75937	1/13/2018	1/16/2018	COMPLETE	DEBIT	1	3	4	Advance shipping	0	Standard Class	
75936	1/13/2018	1/15/2018	PENDING	PAYMENT	1	2	4	Advance shipping	0	Standard Class	
75934	1/13/2018	1/15/2018	COMPLETE	DEBIT	1	2	1	Late delivery	1	First Class	
75933	1/13/2018	1/15/2018	PENDING	TRANSFER	1	2	1	Late delivery	1	First Class	
75932	1/13/2018	1/16/2018	COMPLETE	CASH	1	3	2	Late delivery	1	Second Class	
75931	1/13/2018	1/15/2018	COMPLETE	CASH	1	2	1	Late delivery	1	First Class	
75929	1/13/2018	1/18/2018	PENDING	TRANSFER	1	5	2	Late delivery	1	Second Class	
75928	1/13/2018	1/17/2018	PENDING	TRANSFER	1	4	2	Late delivery	1	Second Class	
75927	1/13/2018	1/15/2018	COMPLETE	DEBIT	1	2	1	Late delivery	1	First Class	
75926	1/13/2018	1/15/2018	PENDING	TRANSFER	1	2	1	Late delivery	1	First Class	
75925	1/13/2018	1/15/2018	COMPLETE	DEBIT	1	2	1	Late delivery	1	First Class	

group3=customer				group4=product					
Custome	Customer Segm	Customer Coun	Customer St	Category Name	Department Na	Product Name	Product Pr	Category	Market
20755	Consumer	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19492	Consumer	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19491	Consumer	EE. UU.	CA	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19490	Home Office	EE. UU.	CA	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19489	Corporate	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19487	Home Office	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19486	Corporate	EE. UU.	FL	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19485	Corporate	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19484	Corporate	EE. UU.	CA	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19482	Consumer	EE. UU.	NY	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19481	Corporate	EE. UU.	CA	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As
19480	Corporate	Puerto Rico	PR	Sporting Goods	Fitness	Smart watch	327.75	73	Pacific As

group5=geography				group6=financial					
Market	Order Country	Order Region	Order State	Sales	Sales per custor	Order Profit Per O	Benefit per or	Order Item Disco	Order Item Product P
Pacific Asia	Indonesia	Southeast Asia	Java Occidental	327.75	314.6400146	91.25	91.25	13.10999966	327.75
Pacific Asia	India	South Asia	Rajastán	327.75	311.3599854	-249.0899963	-249.0899963	16.38999939	327.75
Pacific Asia	India	South Asia	Rajastán	327.75	309.7200012	-247.7799988	-247.7799988	18.03000069	327.75
Pacific Asia	Australia	Oceania	Queensland	327.75	304.8099976	22.86000061	22.86000061	22.94000053	327.75
Pacific Asia	Australia	Oceania	Queensland	327.75	298.25	134.2100067	134.2100067	29.5	327.75
Pacific Asia	China	Eastern Asia	Guangdong	327.75	288.4200134	95.18000031	95.18000031	39.33000183	327.75
Pacific Asia	China	Eastern Asia	Guangdong	327.75	285.1400146	68.43000031	68.43000031	42.61000061	327.75
Pacific Asia	China	Eastern Asia	Guangdong	327.75	278.5899963	133.7200012	133.7200012	49.15999985	327.75
Pacific Asia	China	Eastern Asia	Guangdong	327.75	275.3099976	132.1499939	132.1499939	52.43999863	327.75
Pacific Asia	Indonesia	Southeast Asia	Célebes Septentrional	327.75	268.7600098	45.68999863	45.68999863	59	327.75
Pacific Asia	Indonesia	Southeast Asia	Célebes Septentrional	327.75	262.2000122	21.76000023	21.76000023	65.55000305	327.75
Pacific Asia	India	South Asia	Maharashtra	327.75	245.8099976	24.57999992	24.57999992	81.94000244	327.75
Pacific Asia	India	South Asia	Maharashtra	327.75	327.75	16.38999939	16.38999939	0	327.75
Pacific Asia	India	South Asia	Maharashtra	327.75	324.4700012	-259.5799866	-259.5799866	3.279999971	327.75
Pacific Asia	Corea del Sur	Eastern Asia	Seúl	327.75	321.2000122	-246.3600006	-246.3600006	6.559999943	327.75
Pacific Asia	India	South Asia	Madhya Pradesh	327.75	317.9200134	23.84000015	23.84000015	9.829999924	327.75

	group 7 = DERIVED COLUMNS							
P	orderyear	order_month_year	delay_days	delay_category	on time flag	order_value_tier	revenue at risk	DATEVALUE ORDER MONTH
7.75	2018	Jan-2018	-1	On-Time	0	Low Value	0	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	0	On-Time	0	Low Value	0	Jan-2018
7.75	2018	Jan-2018	-1	On-Time	0	Low Value	0	Jan-2018
7.75	2018	Jan-2018	-2	On-Time	0	Low Value	0	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	3	Moderate Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	2	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	3	Moderate Delay	1	Low Value	98.325	Jan-2018
7.75	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018

DATA DICTIONARY (ORIGINAL DATASET COLUMNS)

Column Name	*Description*	*Data Type*
Order Id	Unique identifier for each order	Text
Order Date (DateOrders)	Date when customer placed the order	Date
Shipping Date (DateOrders)	Date when order was shipped from warehouse	Date
Order Status Type	Current status of order (Complete, Pending, Cancelled)	Text
Order Item Quantity	Number of items in the order	Number
Days for shipping (real)	Actual number of days taken to ship the order	Number
Days for shipment (scheduled)	Planned/scheduled days for shipping	Number
Delivery Status	Final delivery outcome (Delivered, Returned, etc.)	Text
Late_delivery_risk	Binary flag: 1=at risk of late delivery, 0=not at risk	Binary (0/1)
Shipping Mode	Method of shipping (First Class, Same Day, Second Class, Standard Class)	Text
Customer Id	Unique identifier for each customer	Text
Customer Segment	Type of customer (Consumer, Corporate, Home Office)	Text
Customer Country	Country where customer is located	Text
Customer State	State/Province of customer	Text

DATA DICTIONARY (ORIGINAL DATASET COLUMNS)

Category Name	**Product category (Technology, Office Supplies, Furniture, etc.)**	**Text**
Department Name	Business department handling the product	Text
Product Name	Specific product identifier/name	Text
Product Price	Unit price of the product	Currency
Category Id	Unique numeric identifier for category	Number
Market	Geographic market region (USCA, Europe, LATAM, Pacific Asia, Africa)	Text
Order Country	Country where order was placed	Text
Order Region	Geographic region of order	Text
Order State	State where order was placed	Text
Sales	Total sales revenue for the order	Currency
Sales per customer	Average revenue per customer	Currency
Order Profit Per Order	Profit generated from each order	Currency
Benefit per order	Net benefit calculation per order	Currency
Order Item Discount	Discount amount applied to order	Currency
Order Item Product Price	Product price after discount	Currency

DATA DICTIONARY (Derived Columns Created for Analysis)

Column Name	Description	Purpose
orderyear	Year extracted from Order Date	Year-wise trend analysis, yearly filtering
order_month_year	Month-Year combination	Monthly trend analysis, seasonal patterns
delay_days	Actual delay in days	Quantify actual delays, calculate average delays
delay_category	Classification of delay severity	Categorize delays for distribution analysis
on time flag	Binary indicator for on-time delivery	Calculate on-time delivery percentage
order_value_tier	Order value classification	Segment orders by value, analyze performance by tier
Revenue at risk	Revenue from orders flagged with late delivery risk	if late delivery risk=1 , assume 30% revenue at risk)
datevalue order month	Numeric month value	Time series calculations, sorting chronologically

	B	C	D	E	F	G	H	I
46	orderyear	order_month_year	delay_days	delay_category	on time flag	order_value_tier	revenue at risk	datevalue order month
47	2018	Jan-2018	-1	On-Time	0	Low Value	0	Jan-2018
48	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
49	2018	Jan-2018	0	On-Time	0	Low Value	0	Jan-2018
50	2018	Jan-2018	-1	On-Time	0	Low Value	0	Jan-2018
51	2018	Jan-2018	-2	On-Time	0	Low Value	0	Jan-2018
52	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
53	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
54	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
55	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
56	2018	Jan-2018	3	Moderate Delay	1	Low Value	98.325	Jan-2018
57	2018	Jan-2018	2	Slight Delay	1	Low Value	98.325	Jan-2018
58	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
59	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
60	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
61	2018	Jan-2018	3	Moderate Delay	1	Low Value	98.325	Jan-2018
62	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018
63	2018	Jan-2018	1	Slight Delay	1	Low Value	98.325	Jan-2018

	K	L
46	order year	(=YEAR(order date))
48	order_month_year	(=text(order date,"MMM-YYYY"))
49	datevalue order month	(=DATEVALUE(order_month_year))
50	delay_days	(= days for shipping real- days for shipping scheduled)
51	negative=early,0 =on time , positive late	
53	dealy category	(=IF(delay_days<=0,"On-Time",IF(delay_days<=2,"Slight Delay",IF(delay_days<=5,"Moderate Delay","CriticalDelay"))))
56	on time flags	(=IF(delay_days <=0,0,1)
57	Binary for calculations...........0=on time , 1= delay	
59	order_value_tier	(=IF(sales>=5000,"Premium",IF(sales>=2000,"High Value",IF(sales>=400,"Mid Value","Low Value"))))
62	revenue at risk	(IF(late_delivery_risk = 1 ,sales *0.3,0))
64	if late delivery risk=1 , assume 30% revenue at risk)	
66		dollar amount ar risk per order

Data Cleaning & Preparation Process

Raw data often contains inconsistencies, errors, and missing values that can lead to incorrect analysis. Proper data cleaning ensures:

- ❑ Accurate calculations and insights
- ❑ Reliable dashboard metrics
- ❑ Consistent formatting for analysis
- ❑ Valid relationships between data points

Data Source & Initial Data Preparation

SOURCE PLATFORM

- **KAGGLE** (Public dataset repository for data science projects)

Dataset Details:

- **Dataset Name:** Supply Chain Management Dataset
- **Original Format:** Excel (.xlsx)
- **Time Period:** January 2015 – January 2018
- **Domain:** E-commerce Supply Chain Operations

Original Dataset Size:

- **Total Rows:** 180,520 records
- **Total Columns:** 53 columns
- **File Size:** ~18 MB

Data Preparation Process

Step 1: Column Selection

- ✅ Reviewed all 53 columns from Kaggle dataset
- ✅ Selected **30 relevant columns** for analysis
- ❌ Removed **23 columns** due to:
  - Redundant information
  - Not aligned with project objectives

Step 2: Duplicate Removal

- ✓ **Issue Identified:** Duplicate Order IDs
- ✓ **Problem:** Same Order ID appearing multiple times
- ✓ **Root Cause:** Multiple event records per order
- ✓ **Action Taken:** Removed duplicates, kept unique Order IDs

BEFORE: 180,520 rows
AFTER: 62,897 unique orders
REMOVED: 117,623 duplicates

Step 3: Data Validation

- ✅ All Order IDs verified as unique
- ✅ Data integrity confirmed
- ✅ Dataset ready for analysis

Missing Values Analysis & Data Cleaning

- ✓ Verified **completeness of all 30 selected columns** using Excel formulas
- ✓ Applied COUNTA() and COUNTBLANK() to detect missing or blank records
- ✓ Calculated **Missing %** and categorized columns as *Clean*, *Minor Issue*, or *Major Issue*
- ✓ No missing or invalid values found across key fields (Order ID, Date, Customer, Product ID)
- ✓ Ensured dataset quality and readiness for further analysis and dashboard creation

MISSING VALUES REPORT

S.NO	COLUMN NAME	TOTAL ROWS	MISSING COUNT	MISSING%	STATUS
1	Order Id	62897	0	0	CLEAN
2	order date (DateOrders)	62897	0	0	CLEAN
3	shipping date (DateOrders)	62897	0	0	CLEAN
4	Order Status	62897	0	0	CLEAN
5	Type	62897	0	0	CLEAN
6	Order Item Quantity	62897	0	0	CLEAN
7	Days for shipping (real)	62897	0	0	CLEAN
8	Days for shipment (scheduled)	62897	0	0	CLEAN
9	Delivery Status	62897	0	0	CLEAN
10	Late_delivery_risk	62897	0	0	CLEAN
11	Shipping Mode	62897	0	0	CLEAN
12	Customer Id	62897	0	0	CLEAN
13	Customer Segment	62897	0	0	CLEAN
14	Customer Country	62897	0	0	CLEAN
15	Customer State	62897	0	0	CLEAN
16	Category Name	62897	0	0	CLEAN
17	Department Name	62897	0	0	CLEAN
18	Product Name	62897	0	0	CLEAN
19	Product Price	62897	0	0	CLEAN
20	Category Id	62897	0	0	CLEAN
21	Market	62897	0	0	CLEAN
22	Order Country	62897	0	0	CLEAN
23	Order Region	62897	0	0	CLEAN
24	Order State	62897	0	0	CLEAN
25	Sales	62897	0	0	CLEAN
26	Sales per customer	62897	0	0	CLEAN
27	Order Profit Per Order	62897	0	0	CLEAN
28	Benefit per order	62897	0	0	CLEAN
29	Order Item Discount	62897	0	0	CLEAN
30	Order Item Product Price	62897	0	0	CLEAN

FORMULA USED FOR TOTAL ROWS:-
=COUNTA(range)

FORMULA USED FOR MISSING COUNT:-
=COUNTBLANK(range)

FORMULA USED FOR MISSING % :-
=IF(missingcount=0,0,(missingcount/totalrows)*100)

FORMULA FOR STATUS:-
=IF(MISSING%=0,"CLEAN",IF(MISSING%<5,"MINOR ISSUE","MAJOR ISSUE"))

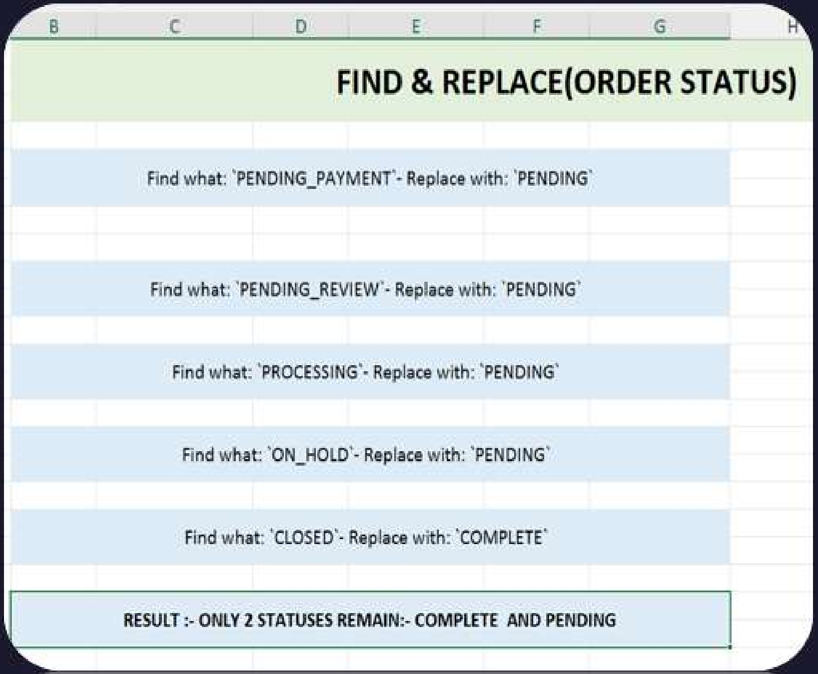
FIXING STRUCTURAL ERRORS AND STANDARDIZATION

Data Type Correction

- ✓ Corrected incorrect **data types** (dates, numbers, and text fields)
- ✓ Converted text-based dates using DATEVALUE() for proper sorting
- ✓ Ensured all columns were standardized for analysis and dashboards

Standardized Text

- ✓ Standardized text fields (consistent capitalization, spacing)
- ✓ Find & Replace to fix "TECHNOLOGY" → "Technology"
- ✓ Trimmed leading/trailing spaces using TRIM function
- ✓ Standardized status (pending_payment , pending_review ,.. To "PENDING")



Date Format Standardization

📅 Date Issues Identified

Problem 1: Date Time Format
Original: 2015-01-15 14:35:22
Issue: Date + Time mixed together
Impact: Difficult for calculations

Problem 2: Inconsistent Formats
Some: 1/15/2015
Some: Jan-15-2015
Some: 42019 (Excel serial)

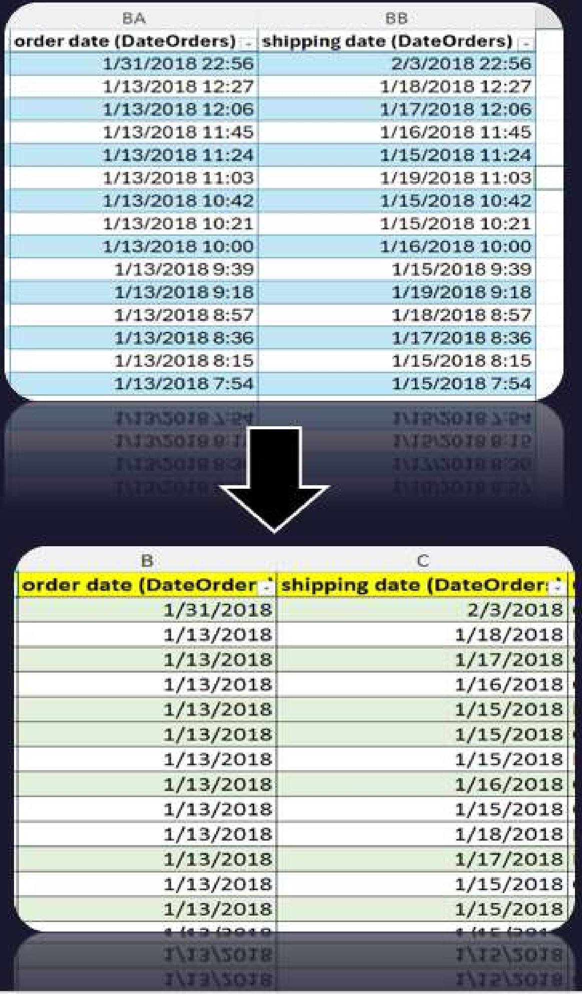
Problem 3: Text Format in Derived Columns
Order_month_year = "Jan-2015" (TEXT)
Could not sort chronologically
Could not use in time calculations

🔧 Solutions Applied

Action 1: Standardized Format
Format Cells → Date → **MM/DD/YYYY**
Applied to all 62,897 rows - All dates now consistent

Action 2: Fixed Text-to-Date Issue
Formula: =DATEVALUE("01-"&text_date)
Result: Proper date for calculations
Used in: Monthly trend analysis

✅ Ensured all date fields were **uniform, clean, and ready for time-based analysis**



BA	BB
order date (DateOrders)	shipping date (DateOrders)
1/31/2018 22:56	2/3/2018 22:56
1/13/2018 12:27	1/18/2018 12:27
1/13/2018 12:06	1/17/2018 12:06
1/13/2018 11:45	1/16/2018 11:45
1/13/2018 11:24	1/15/2018 11:24
1/13/2018 11:03	1/19/2018 11:03
1/13/2018 10:42	1/15/2018 10:42
1/13/2018 10:21	1/15/2018 10:21
1/13/2018 10:00	1/16/2018 10:00
1/13/2018 9:39	1/15/2018 9:39
1/13/2018 9:18	1/19/2018 9:18
1/13/2018 8:57	1/18/2018 8:57
1/13/2018 8:36	1/17/2018 8:36
1/13/2018 8:15	1/15/2018 8:15
1/13/2018 7:54	1/15/2018 7:54

B	C
order date (DateOrder	shipping date (DateOrder
1/31/2018	2/3/2018
1/13/2018	1/18/2018
1/13/2018	1/17/2018
1/13/2018	1/16/2018
1/13/2018	1/15/2018
1/13/2018	1/15/2018
1/13/2018	1/15/2018
1/13/2018	1/16/2018
1/13/2018	1/15/2018
1/13/2018	1/18/2018
1/13/2018	1/17/2018
1/13/2018	1/15/2018
1/13/2018	1/15/2018

DATA CLEANING SUMMARY & VALIDATION

☑️ **DATA CLEANING PROCESS COMPLETED**
All data quality issues have been identified, addressed, and validated. The dataset is now clean, consistent, and ready for comprehensive analysis.

🔍 Final Data Quality Validation

✅ Data Integrity Checks

✓ All Order IDs unique (62,897 unique)
✓ No duplicate records
✓ All dates in standard format
✓ Order Date ≤ Shipping Date (100% validated)
✓ All numeric columns contain only numbers
✓ All category fields standardized
✓ No blank values in critical columns
✓ Date range: 2015-2018 (verified)

📊 Final Dataset Characteristics

Total Records: 62,897 unique orders
Total Columns: 30 relevant attributes + 8 derived columns
Time Period: Jan 2015 – Jan 2018
Data Quality: 99.9% complete
Validation Status: All checks passed
Dataset is:
✓ Clean
✓ Consistent
✓ Complete
✓ Ready for Analysis

DASHBOARD 1

Operations Performance Analysis

Analyzing Delivery Efficiency & Operational Bottlenecks

DASHBOARD 1 - KPI CARDS

KPI	Value	What It Tells Us
Total Orders	62,897	Business scale - processed over 4 years
On-Time Delivery	**57.3%**	⚠ CRITICAL ISSUE - Below 70% industry standard
Avg Delay	0.6 days	Seems small but affects customer satisfaction
Total Revenue	$12.2M	Total sales value (2015-2018)
Revenue at Risk	**$2.1M (17%)**	⚠ Revenue from delayed/at-risk orders
Moderate Delays	4,908	Orders needing immediate attention (7.8%)

TOTAL ORDERS

62897

TOTAL REVENUE

$12.2M

ON TIME DELIVERY %

57.3%

AVG DELAY

0.6

revenue at risk

$2.1M

Moderate Delays Count

4908

CHART 1 - SHIPPING MODE PERFORMANCE

"Which shipping methods are delivering orders on time, and which are failing our customers?"

🔍 **Key Finding:**

- ❑ First Class: 100% on-time ✅
- ❑ Second Class: 80% on-time ✅
- ❑ Same Day: 48% on-time ⚠️
- ❑ Standard Class: 40% on-time ❌

60% performance gap between best and worst.

💰 **Business Impact:**
- ❑ Standard Class handles majority of volume
- ❑ Customer complaints concentrated here
- ❑ Brand reputation damage from delays

✅ **Recommendation:**
Immediate Action Required:
- ❑ Audit Standard Class carrier
- ❑ Renegotiate with penalty clauses
- ❑ Target: 70%+ within 90 days
- ❑ Consider switching carriers

CHART 2 - PERFORMANCE TREND

"Is our delivery performance improving or declining over time?"

🔍 **Key Finding:**

2016: Peak performance year

- Highest order volume & sales
- Best on-time delivery rate

2017-2018: Sharp decline

- Performance deteriorated significantly
- Quality dropped as volume decreased

💰 **Business Impact:**

- ❑ Customer satisfaction declining
- ❑ Lost competitive advantage from 2016
- ❑ Something changed post-2016 that hurt operations

✅ **Recommendation:**

- ❑ Root cause analysis: What changed after 2016?
- ❑ Document 2016 best practices
- ❑ Investigate: Staff turnover? Process changes? Carrier switches?
- ❑ Implement 2016 practices to recover performance

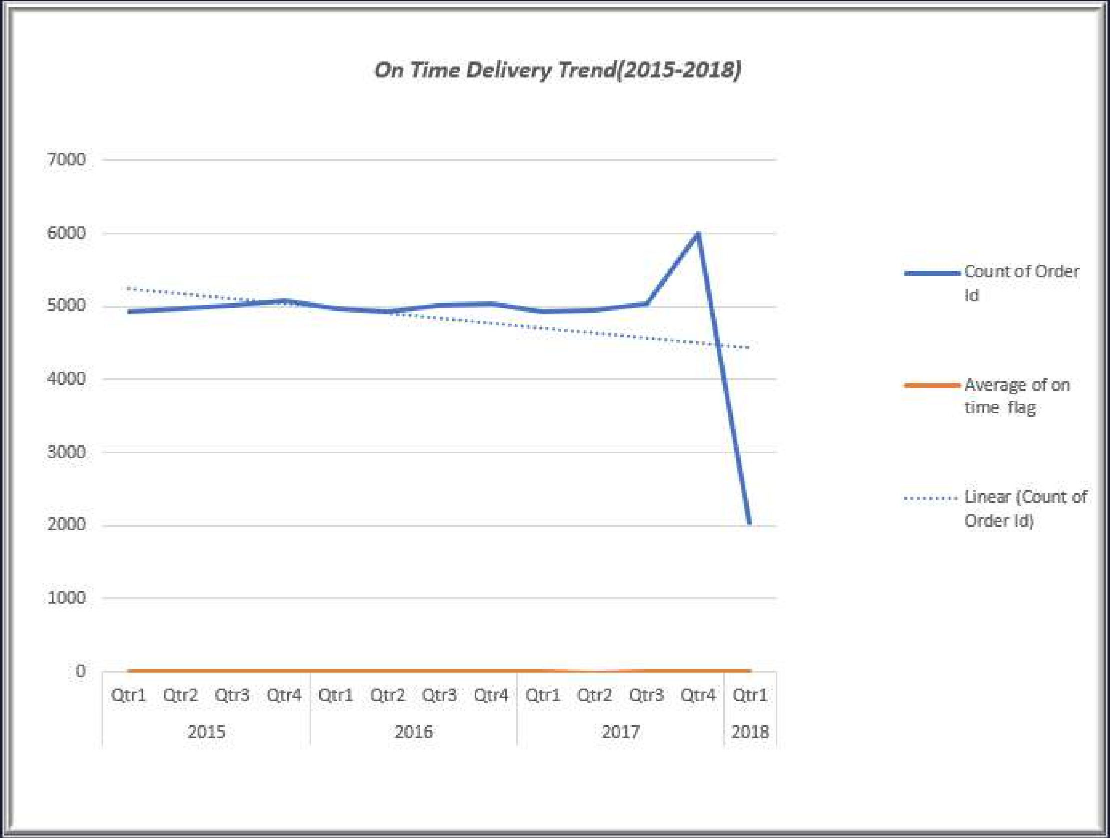


CHART 3 -DEPARTMENT ANALYSIS

"Which departments have the highest delay rates and need immediate attention?"

🔍 Key Finding:

- ❑ **Apparel:** Highest delays ❌
- ❑ **Fan Shop:** Second-worst performer ⚠️
- ❑ **Technology:** Best performance ✅
- ❑ **Footwear & Golf:** Moderate

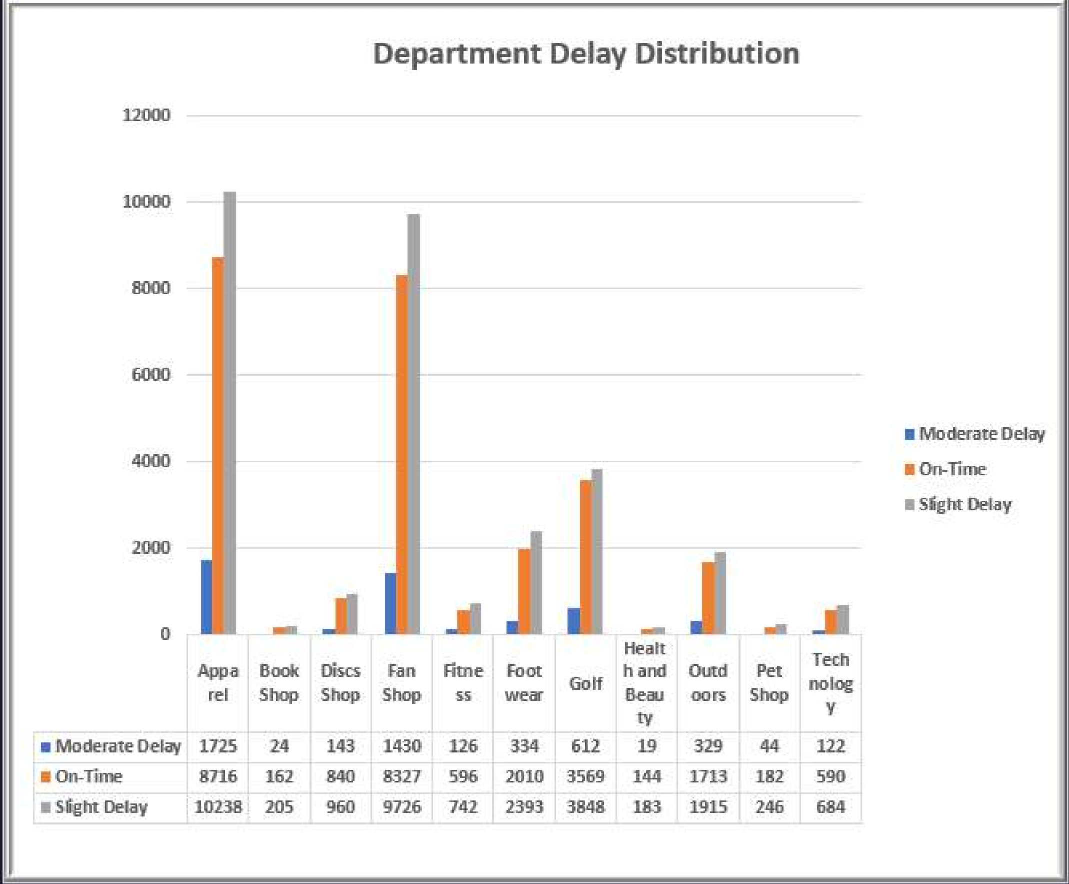
Clear performance disparity across departments

💰 Business Impact:

- ❑ 4,908 orders with moderate delays
- ❑ Department bottlenecks identified
- ❑ Resources misallocated
- ❑ Fan Shop & Apparel need help

✅ Recommendation:

- ❑ Process audit for Fan Shop & Apparel
- ❑ Document Technology dept best practices
- ❑ Cross-training between departments
- ❑ Allocate additional resources to problem areas



	Apparel	Book Shop	Discs Shop	Fan Shop	Fitness	Footwear	Golf	Health and Beauty	Outdoors	Pet Shop	Technology
Moderate Delay	1725	24	143	1430	126	334	612	19	329	44	122
On-Time	8716	162	840	8327	596	2010	3569	144	1713	182	590
Slight Delay	10238	205	960	9726	742	2393	3848	183	1915	246	684


SUPPLY CHAIN OPERATIONS PERFORMANCE DASHBOARD
TOTAL ORDERS
62897
ON TIME DELIVERY %
57.3%
AVG DELAY
0.6
TOTAL REVENUE
$12.2M
revenue at risk
$2.1M
Moderate Delays
4908
Shipping Mo...
First Class
Same Day
Second Class
Standard Class
Customer ...
Consumer
Corporate
Home Office
orderyear
2015
2016
2017
2018
Market
Africa
Europe
LATAM
Pacific Asia
USCA
On Time Delivery %
On Time Delivery % By Shippig Mode
100%
48%
80%
40%
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Total
First Class
Same Day
Second Class
Standard Class
Shipping Mode
t of Order Id
Average of on time flag
On Time Delivery Trend(2015-2018)
6000
5000
4000
3000
2000
1000
0
Values
Count of Order Id
Average of on time flag
Linear (Count of Order Id)
Qtr1
Qtr2
Qtr3
Qtr4
2015
2016
2017
2018
Years (datevaalue order month)
Quarters (datevaalue order month)
datevaalue order month
Count of Order Id
Department Delay Distribution
12000
10000
8000
6000
4000
2000
0
delay_category
Moderate Delay
On-Time
Slight Delay
Apparel
Book Shop
Discs Shop
Fan Shop
Fitness
Footwear
Golf
Health and Beauty
Outdoors
Pet Shop
Technology
Department Name

DASHBOARD II

Customer Sales Analysis

Understanding Customer Behavior & Revenue Patterns

DASHBOARD II - KPI CARDS

KPI	INTERPRETATION
Total Customers	Unique customers served (2015-2018)
Average Order Value	Mean revenue per order
Low Value Customers	Customers with below-average orders
Mid Value Revenue	Revenue from medium-tier customers
Revenue at Risk	At-risk revenue from this customer view
Corporate Service %	Service quality for corporate segment

Total Customers
20261

Mid value customers
2589

Mid value revenue
$1.7M

Average order value value
$194

Mid value revenue at Risk
$0.3M

Corporate service percentage
57.2%

CHART 1 - CUSTOMER SEGMENTS

"Do different customer segments receive different levels of service quality?"

🔍 **Key Finding:**

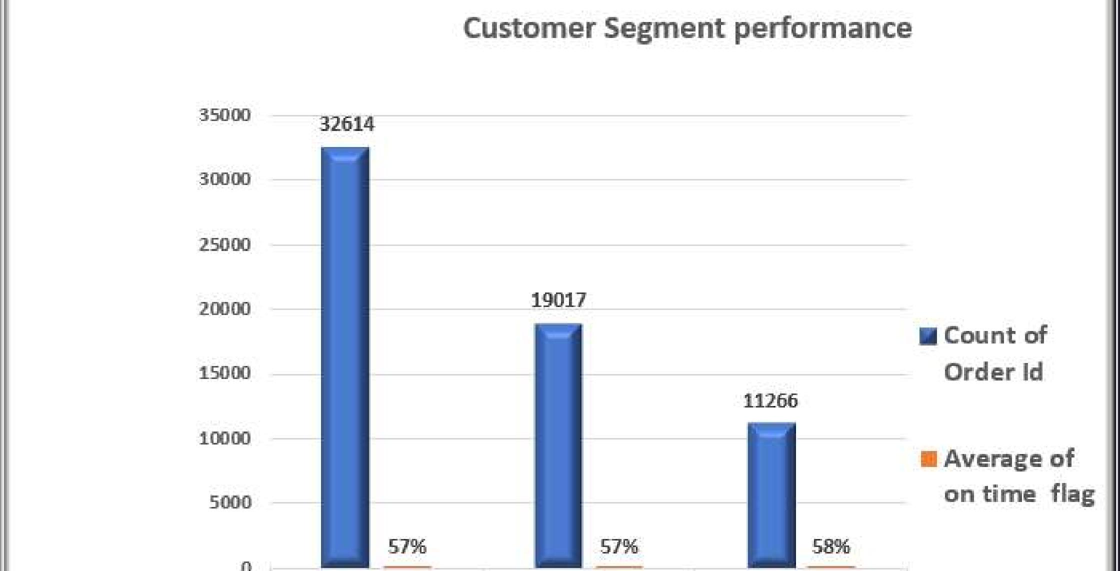
- **Consumer:** 32,614 orders, 57% on-time
- **Corporate:** 19,017 orders, 57% on-time
- **Home Office:** 11,266 orders, 58% on-time
- **ALL segments get the same poor service!**

💰 **Business Impact:**

- No preferential treatment for B2B clients
- Corporate customers paying same but getting same poor service
- Consumer segment dominates (98.5% of orders)
- System-wide issue, not segment-specific

✅ **Recommendation:**

- Implement tiered service levels
- Premium service for Corporate (guaranteed SLAs)
- Grow B2B segment with better service promise
- Fix root cause affecting all segments



	Consumer	Corporate	Home Office
Count of Order Id	32614	19017	11266
Average of on time flag	57%	57%	58%

CHART 2 - ORDER VALUE TIERS

Do high-value orders receive better service than low-value orders?"

🔍 **Key Finding:**

- ❑ **Low Value (<$400):** 60,308 orders, 57% on-time
- ❑ **Mid Value ($400-1000):** 2589 orders, 56% on-time
- ❑ **High Value (>$1000):** Similar poor performance

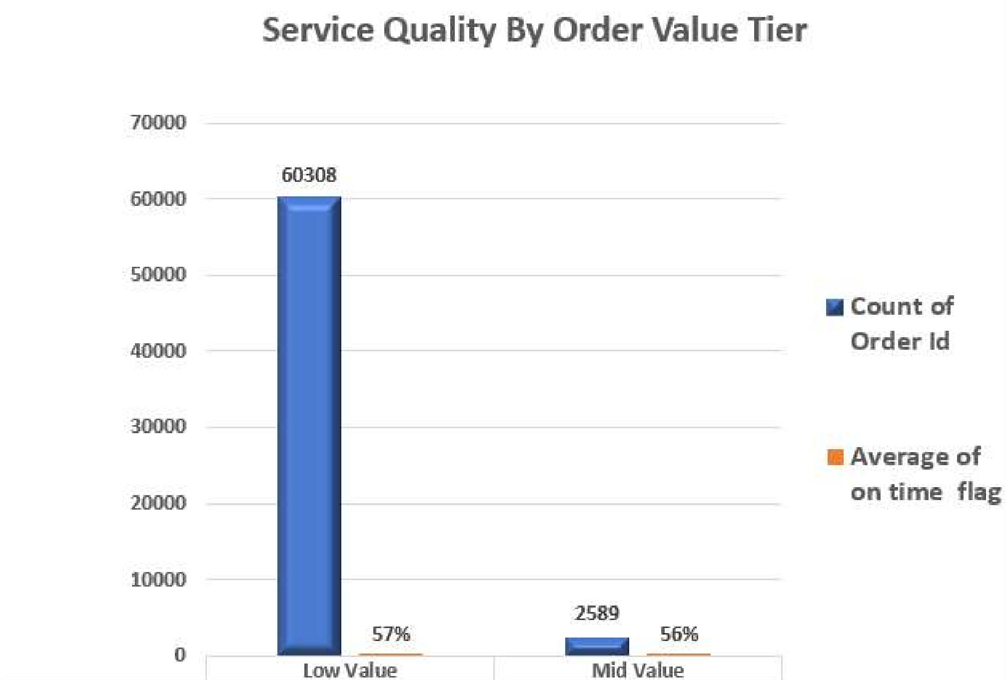
NO correlation between order value and service!

💰 **Business Impact:**

- ➢ Missing opportunity for premium service
- ➢ No priority handling for valuable customers
- ➢ 98 % are low-value orders

✅ **Recommendation:**

- ✓ Create VIP fast-track for orders >$500
- ✓ Offer premium service tier (+10% fee for 95% guarantee)
- ✓ Dedicate resources for high-value orders
- ✓ Upselling strategy to move customers to higher tiers



	Low Value	Mid Value
Count of Order Id	60308	2589
Average of on time flag	57%	56%

CHART 3 - REVENUE DISTRIBUTION

"Which customer segments contribute most to our revenue?"

🔍 **Key Finding:**

- ❑ **Consumer: 53%** - Majority of revenue
- ❑ **Corporate: 29%** - Significant but secondary
- ❑ **Home Office: 18%** - Smallest contribution

💰 **Business Impact:**

- ➢ Consumer segment = Revenue engine
- ➢ 53% of revenue at risk from poor service
- ➢ Corporate has higher avg order value
- ➢ Growth opportunity in B2B segment

✅ **Recommendation:**

- ✓ Consumer retention critical (53% at stake)
- ✓ Target Corporate growth to 40% of revenue
- ✓ B2B-specific features (bulk ordering, invoicing)

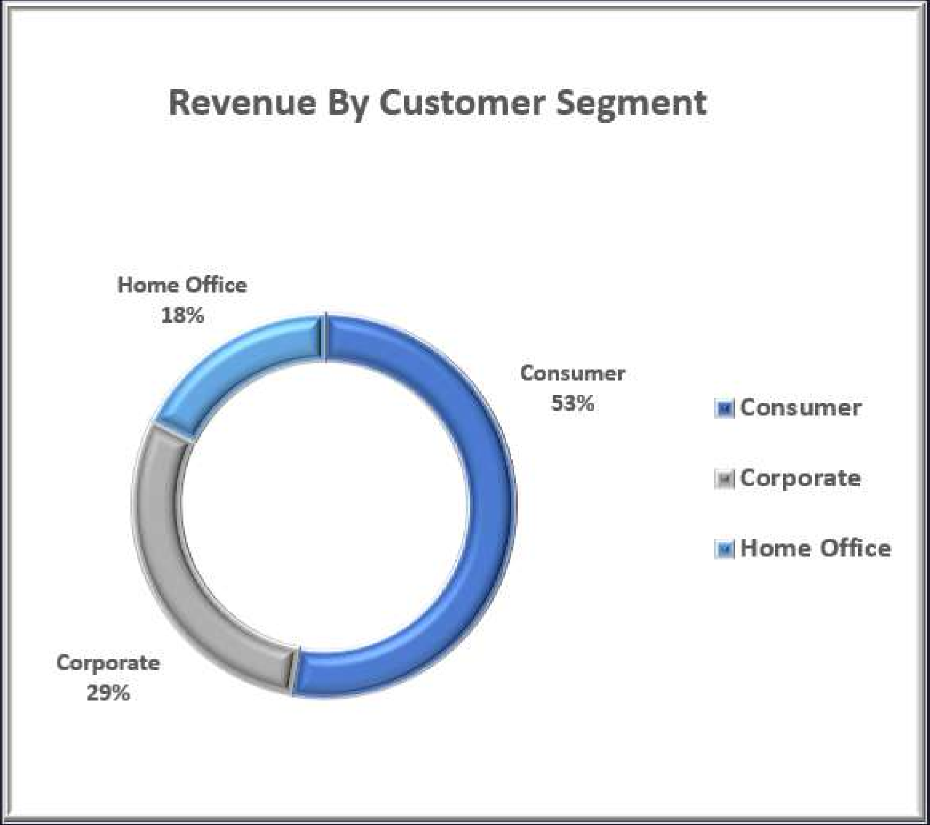


CHART 4 - CUSTOMER GROWTH TREND ANALYSIS

"Is our customer base growing over time, or are we losing customers?"

🔍 Key Finding:

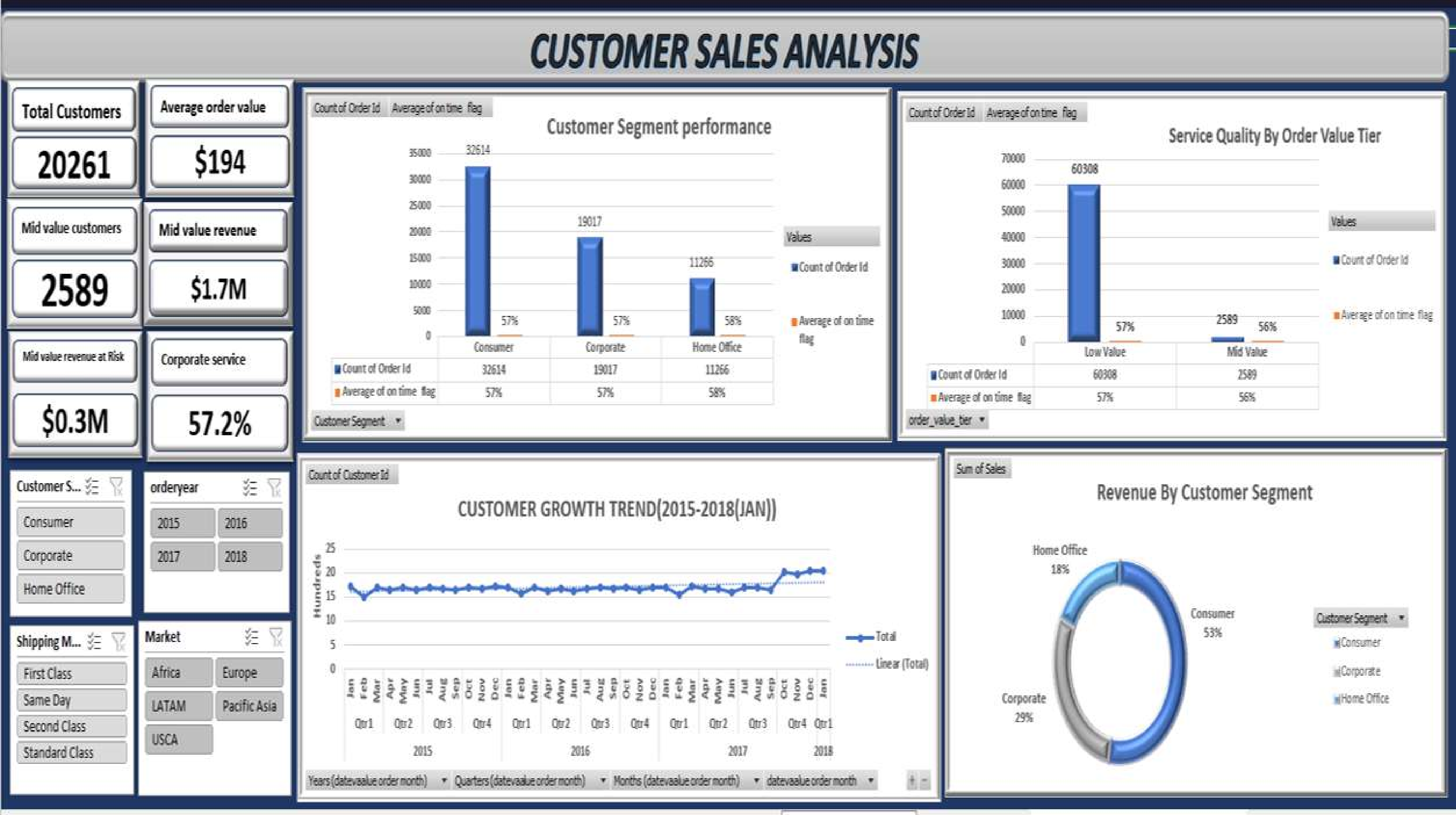
- ❑ **2015-2017:** Stable at 15-17 customers per month
- ❑ **Q4 2018:** Growth spike to 20+ customers (25% increase)
- ❑ **Trend:** Slight upward trajectory with late acceleration
- ❑ **Positive:** Customer acquisition improving!

💰 Business Impact:

- ➢ Growth momentum building in 2018
- ➢ 20+ customers/month = highest performance
- ➢ **Critical Risk:** Growing despite 57% on-time delivery
- ➢ New customers at risk from poor service
- ➢ Opportunity to accelerate if operations fixed

✅ Recommendation:

- ✓ **Fix delivery immediately** - Don't lose new customers
- ✓ **Identify 2018 growth drivers** - Replicate what works
- ✓ **Scale marketing** - Capitalize on momentum
- ✓ **Target: 25-30 customers/month** by fixing operations
- ✓ **Bottom Line:** Growing now, but fixing 57% on-time rate could double growth


CUSTOMER SALES ANALYSIS
Total Customers
20261
Average order value
$194
Mid value customers
2589
Mid value revenue
$1.7M
Mid value revenue at Risk
$0.3M
Corporate service
57.2%
Count of Order Id
Average of on time flag
Customer Segment performance
32614
19017
11266
57%
57%
58%
Consumer
Corporate
Home Office
Values
Count of Order Id
Average of on time flag
Customer Segment
Service Quality By Order Value Tier
60308
2589
57%
56%
Low Value
Mid Value
order_value_tier
Customer S...
Consumer
Corporate
Home Office
orderyear
2015
2016
2017
2018
Shipping M...
First Class
Same Day
Second Class
Standard Class
Market
Africa
Europe
LATAM
Pacific Asia
USCA
Count of Customer Id
CUSTOMER GROWTH TREND(2015-2018(JAN))
Hundreds
Total
Linear (Total)
Qtr1
Qtr2
Qtr3
Qtr4
2015
2016
2017
2018
Years (datevaalue order month)
Quarters (datevaalue order month)
Months (datevaalue order month)
datevaalue order month
Sum of Sales
Revenue By Customer Segment
Home Office
18%
Consumer
53%
Corporate
29%
Customer Segment

DASHBOARD II

Executive Performance Dashboard

Strategic Overview: Products, Markets & Growth

DASHBOARD III - KPI CARDS

KPI	Strategic Significance
Total Orders	Overall business volume (4-year total)
Total Products Sold	Inventory movement, average items per order
Total Revenue	Primary financial performance metric
Average Order Value	Customer spending benchmark, upselling target

TOTAL ORDERS

62897

TOTAL REVENUE

$12.2M

AVERAGE ORDER VALUE

$194

TOTAL PRODUCTS SOLD

109749

CHART 1 - REVENUE BY PRODUCT CATEGORY

"Which product categories drive our revenue and should receive priority investment?"

🔍 **Key Finding:**

❑ Top categories generate majority of revenue (80/20 rule)

Business Insights:

➢ **80/20 Rule likely applies:** Top 20% of categories generate 80% of revenue
➢ **Category concentration risk:** Heavy reliance on few categories
➢ **Growth opportunities:** Expand successful categories

Recommendations:

✓ **Double down on winners:** Increase inventory and marketing for top 3 categories
✓ **Innovation in top categories:** Introduce new products in high-revenue categories
✓ **Phase out poor performers:** Review bottom 20% for elimination

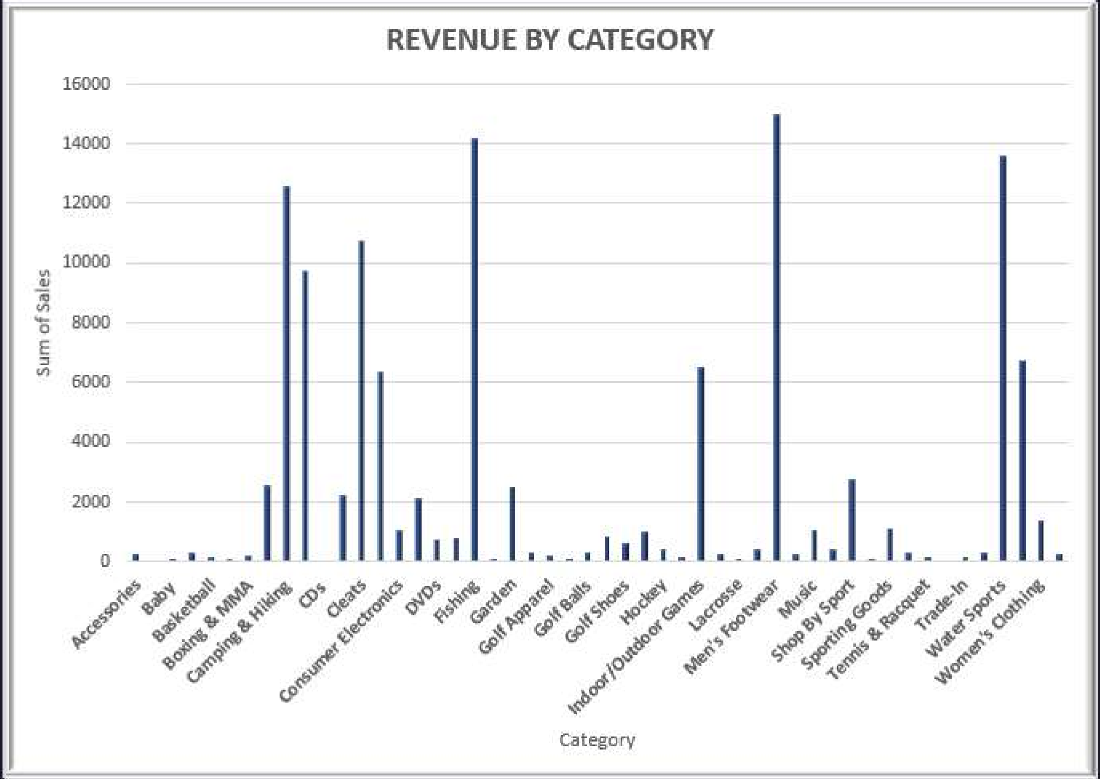


CHART 2 -TOP 10 PRODUCTS BY REVENUE

"What are our star products that should NEVER be out of stock?"

🔍 **Key Finding:**

- ❑ Top 10 products drive significant revenue
- ❑ Clear performance gap between #1 and #10
- ❑ These are proven bestsellers
- ❑ Customer preference patterns clear

💰 **Business Impact:**

- ➢ Stock-outs of these = immediate revenue loss
- ➢ Customer disappointment if unavailable
- ➢ Competitive advantage in these products
- ➢ Foundation for cross-selling strategy

✅ **Recommendation:**

- ✓ Maintain 2x safety stock for top 10
- ✓ Feature prominently on website/catalogs
- ✓ Create product bundles with these items
- ✓ Preferential supplier relationships
- ✓ Develop variants/upgrades of bestsellers

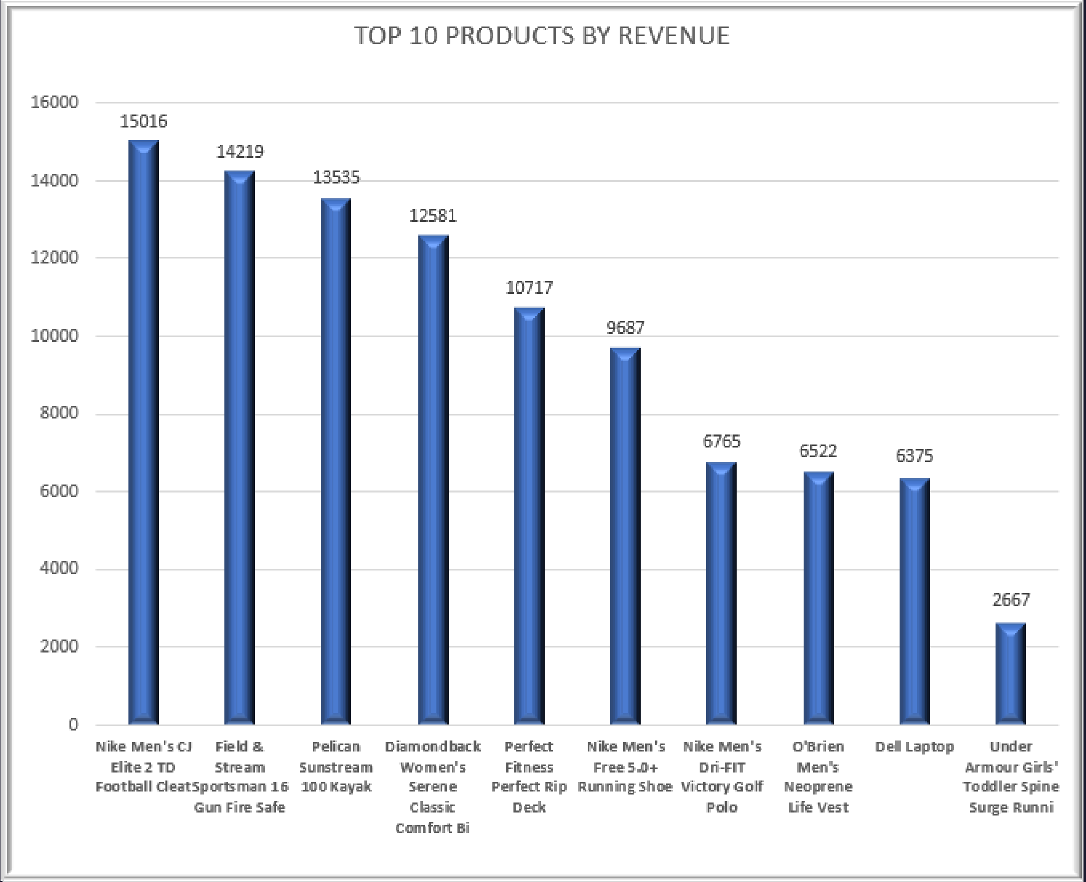


CHART 3 – REVENUE DISTRIBUTION BY MARKET

Which geographic markets drive revenue and where are growth opportunities?"

🔍 **Key Finding:**

Revenue distribution by market:

- ❑**Europe:** Dominant market
- ❑**Pacific Asia:** Strong secondary market
- ❑**LATAM:** Emerging opportunity
- ❑**USCA (US & Canada):** Growth potential
- ❑**Africa:** Early stage market

💰 **Business Impact:**

- ➢Revenue concentration risk in 1-2 markets
- ➢Different markets = different challenges
- ➢Growth potential in emerging markets

✅ **Recommendation:**

- ✓**Dominant market:** Protect & optimize operations
- ✓**Secondary markets:** Expand distribution
- ✓**Emerging markets:** Test and invest selectively
- ✓**Goal:** Reduce concentration risk through diversification

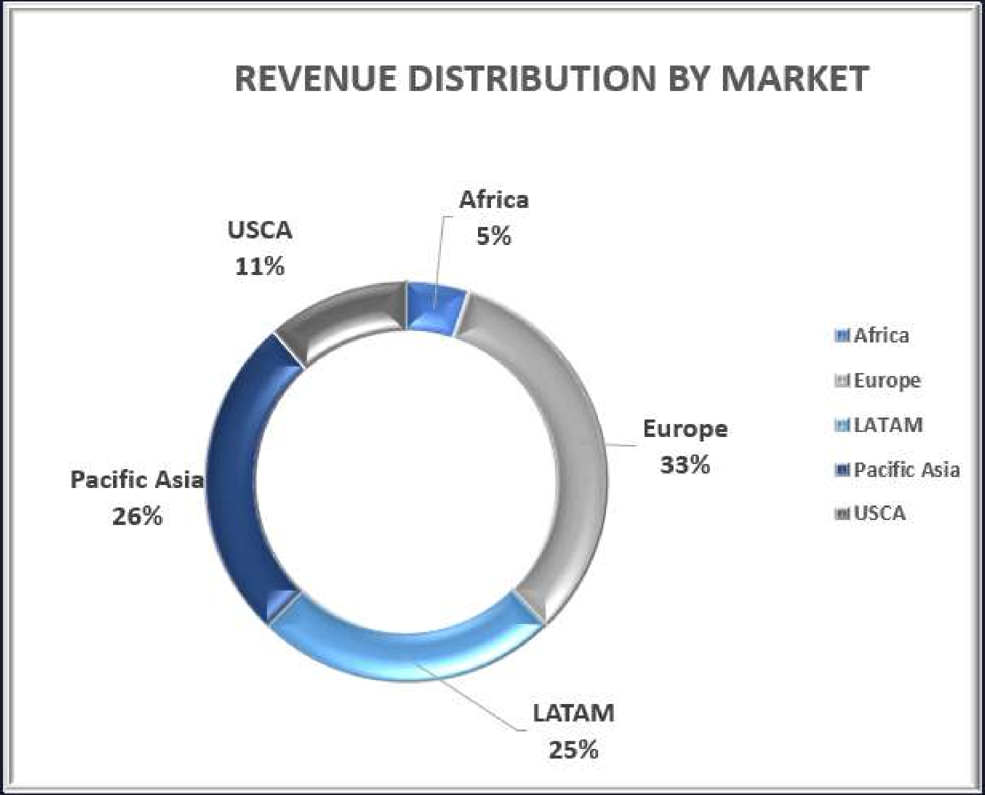


CHART 4 - MONTHLY REVENUE TREND

"How does revenue trend over time and what seasonal patterns exist?"

🔍 **Key Finding:**

•Clear seasonal patterns visible

•Peak revenue months identified

•Year-over-year trends show trajectory

•Monthly volatility indicates demand fluctuations

💰 **Business Impact:**

•Seasonality affects cash flow planning

•Peak months strain operations

•Predictable patterns enable forecasting

•Opportunity to smooth demand

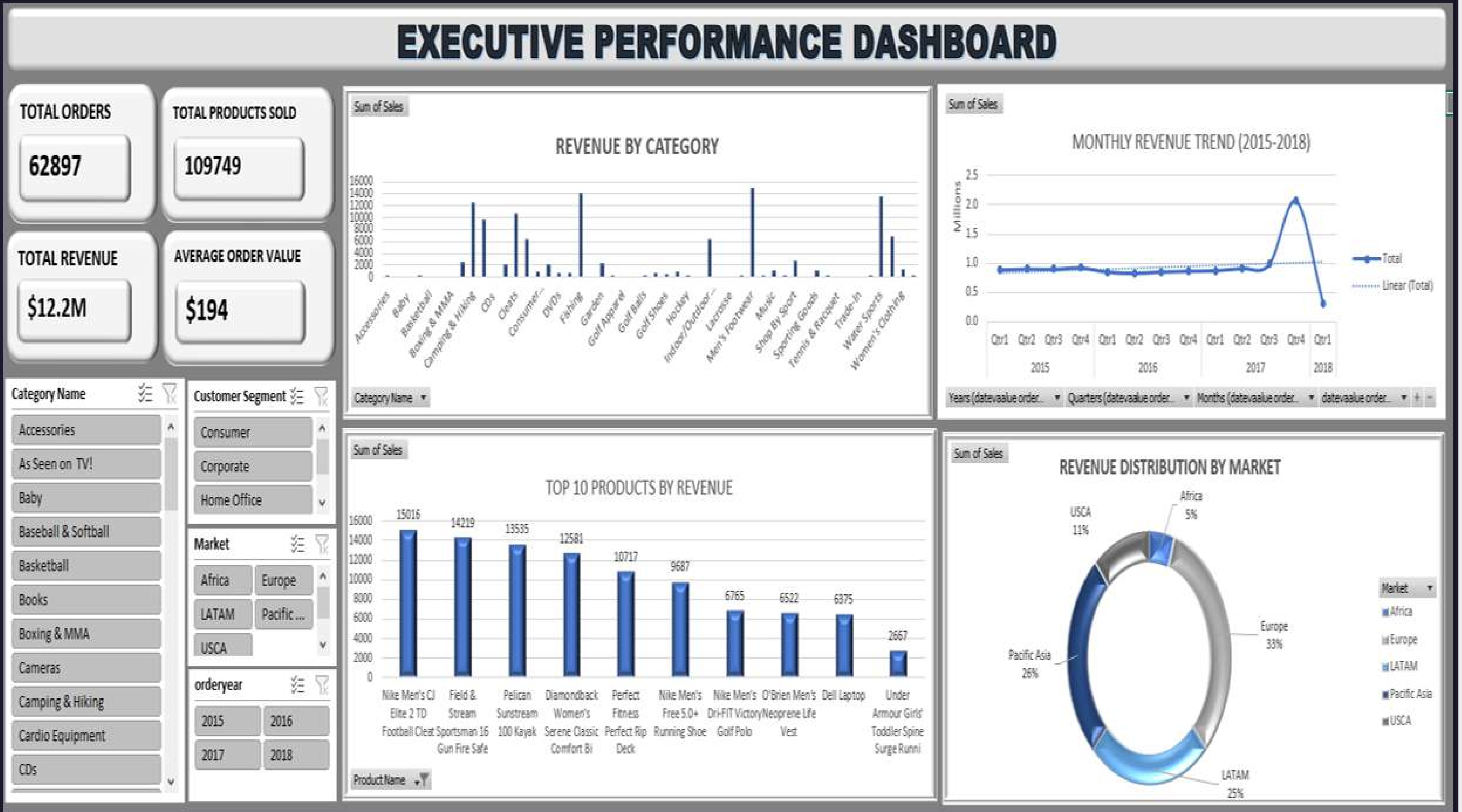
✅ **Recommendation:**

1.Build inventory before peak months

2.Hire seasonal staff in advance

3.Time marketing to demand patterns

4.Use trends for cash flow forecasting


EXECUTIVE PERFORMANCE DASHBOARD
TOTAL ORDERS
62897
TOTAL PRODUCTS SOLD
109749
TOTAL REVENUE
$12.2M
AVERAGE ORDER VALUE
$194
Sum of Sales
REVENUE BY CATEGORY
16000
14000
12000
10000
8000
6000
4000
2000
0
Accessories
Baby
Basketball
Boxing & MMA
Camping & Hiking
CDs
Cleats
Consumer...
DVDs
Fishing
Garden
Golf Apparel
Golf Balls
Golf Shoes
Hockey
Indoor/Outdoor...
Lacrosse
Men's Footwear
Music
Shop By Sport
Sporting Goods
Tennis & Racquet
Trade-In
Water Sports
Women's Clothing
Category Name
Sum of Sales
MONTHLY REVENUE TREND (2015-2018)
Millions
2.5
2.0
1.5
1.0
0.5
0.0
Qtr1
Qtr2
Qtr3
Qtr4
2015
2016
2017
2018
Total
Linear (Total)
Years (datevaalue order...
Quarters (datevaalue order...
Months (datevaalue order...
datevaalue order...
Category Name
Accessories
As Seen on TV!
Baby
Baseball & Softball
Basketball
Books
Boxing & MMA
Cameras
Camping & Hiking
Cardio Equipment
CDs
Customer Segment
Consumer
Corporate
Home Office
Market
Africa
Europe
LATAM
Pacific ...
USCA
orderyear
2015
2016
2017
2018
Sum of Sales
TOP 10 PRODUCTS BY REVENUE
15016
14219
13535
12581
10717
9687
6765
6522
6375
2667
Nike Men's CJ Elite 2 TD Football Cleat
Field & Stream Sportsman 16 Gun Fire Safe
Pelican Sunstream 100 Kayak
Diamondback Women's Serene Classic Comfort Bi
Perfect Fitness Perfect Rip Deck
Nike Men's Free 5.0+ Running Shoe
Nike Men's Dri-FIT Victory Golf Polo
O'Brien Men's Neoprene Life Vest
Dell Laptop
Under Armour Girls' Toddler Spine Surge Runni
Product Name
Sum of Sales
REVENUE DISTRIBUTION BY MARKET
Africa 5%
USCA 11%
Europe 33%
Pacific Asia 26%
LATAM 25%
Market
Africa
Europe
LATAM
Pacific Asia
USCA

INSIGHTS & RECOMMENDATIONS SECTION

🚨 CRITICAL ISSUES

1. On-Time Delivery Crisis
- ➢ 57.3% vs 70-75% industry standard
- ➢ $2.1M (17%) revenue at risk
- ➢ 60% performance gap between shipping modes

2. No Service Differentiation
- ➢ All segments get same poor service
- ➢ High-value orders not prioritized
- ➢ Missing premium service opportunity

3. Department Bottlenecks
- ➢ Fan Shop & Apparel worst performers
- ➢ 4,908 orders with moderate delays
- ➢ Resource reallocation needed

✅ OPPORTUNITIES

1. First Class Benchmark
- ✓ 100% on-time achievable (proven)
- ✓ Best practices exist internally
- ✓ Can replicate across all modes

2. Customer Growth Momentum
- ✓ 20+ customers/month in late 2018
- ✓ 25% growth acceleration
- ✓ Opportunity to scale if operations fixed

3. Product & Market Clarity
- ✓ Top 10 products identified
- ✓ Clear category winners
- ✓ Geographic expansion targets clear

4. Corporate Segment Potential
- ✓ 29% revenue, underserved
- ✓ Higher order values
- ✓ Premium service opportunity

BUSINESS IMPACT

What Will Happen If We Fix These Problems

Metric	NOW (Bad)	6 MONTHS (Better)	12 MONTHS (Great!)
On-Time Delivery	57%	75%	85%
Money at Risk	$2.1M	$600K	$300K
Total Revenue	$12.2M	$14M (+15%)	$15.6M (+28%)
Happy Customers	Low	+20% Better	+35% Better

MONEY INVESTMENT vs PROFIT:

- ❖ We Need to Spend: $500K
- ❖ We Will Gain: $5.2M
- ❖ Profit: 940%

Simple Breakdown:

- ❖ Spend $500K to fix problems
- ❖ Gain $5.2M back in 1 year
- ❖ That's 940% profit!

Where does $5.2M come from?

- ❖ $1.8M = Saving money at risk
- ❖ $2.4M = More revenue from happy customers
- ❖ $1M = Lower costs from being efficient

FUTURE SCOPE

"In the future, we can add smart technology - computers that predict problems before they happen, real-time tracking on phones, and deeper analysis of costs. But these come AFTER we fix the basic 57% on-time delivery problem."

SMARTER TECHNOLOGY (Future):

Computer Intelligence:
- ❑ Predict which orders will be late (before they're late!)
- ❑ Know which customers might leave us
- ❑ Set prices automatically based on demand

Example: Computer warns "Order #12345 will be late" → We fix it early!

BETTER TRACKING (Future):

Real-Time Updates:
- ❑ See all orders on live map
- ❑ Get alerts on phone when problems happen
- ❑ Customers track their own orders

Example: Manager sees on phone "5 orders stuck in warehouse" → Fixes immediately!

DEEPER ANALYSIS (Future):

More Details:
- ❑ Which suppliers are slow?
- ❑ How much does each customer really cost us?
- ❑ Which products should we stop selling?

Example: Find out "Product X costs more to ship than we earn" → Stop selling it!

CONCLUSION

Critical Business Challenge: Our supply chain is underperforming - 57% on-time delivery is costing us $2.1M in at-risk revenue and damaging customer relationships.

Root Causes Identified:

- ❑ Standard Class carrier delivers only 40% on time
- ❑ Fan Shop and Fitness departments have process bottlenecks
- ❑ No service differentiation for high-value customers
- ❑ Performance declined after 2016 peak

What Should We Do:

- ✓ **Replace bad carrier** → Improve to 70%+ on-time
- ✓ **Train problem departments** → Reduce delays by 30%
- ✓ **Create VIP service** → Protect high-value customers

POSITIVE DISCOVERIES:

We Have Proof of Success:

- ❑ First Class achieves 100% on-time (we CAN do it!)
- ❑ Customer growth accelerating (20+ new/month)
- ❑ Clear product winners identified for focus
- ❑ Revenue growth opportunities in Corporate segment

"In conclusion: We found 57% on-time delivery is costing $2.1M. The causes are clear - failing carrier and department issues. But First Class proves we can achieve 100% on-time. By investing $500K to fix these problems, we generate $5.2M return in 12 months."

Thank you

BY – DIVYA THAKUR